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B A N G A L O R E • I N D I A

HarmonyHub- Music Discovery Website

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Introduction

1.1 Purpose

The purpose of this document is to describe the functional and non-functional requirements of **HarmonyHub**, a web-based music discovery platform. HarmonyHub is designed to help users discover songs based on mood, language, genre, artist, and popularity. The platform aims to provide an organized and user-friendly experience for exploring music while allowing registered users to save their favourite songs and playlists in future versions.

1.2 Scope

The scope of this project includes designing and developing a web-based music discovery platform called **HarmonyHub**. The system aims to help users discover songs based on different categories while providing an easy and interactive browsing experience. The proposed website will include the following features:

- Home Page
- Discover Music
- Mood Categories
- Language Categories
- Genre Categories
- Artist Categories
- Trending Songs
- Search Songs
- Login & Registration
- User Dashboard
- Contact Us

1.3 Stack Description

The project uses the following technologies for development:

Frontend: HTML5, CSS, JavaScript

Backend: Node.js and Express.js

Database: MySql

Development Tools: Visual Studio Code

Overall Description

2.1 Requirement

HarmonyHub is developed to provide an interactive digital platform for music lovers who want to discover songs easily. The system allows users to browse music collections, search for songs, and explore different categories without difficulty. Registered users will have access to additional features such as managing favourite song. The application is designed to be simple, responsive, and easy to navigate for users of all ages.

2.2 Module Description

The HarmonyHub web application is divided into several modules:

Home – Displays the homepage with featured and trending songs.

Discover Music – Allows users to explore the complete music collection.

Mood Categories – Displays songs grouped by moods such as Happy, Sad, Romantic, Party, Relaxing, etc.

Language Categories – Allows users to browse songs based on language (English, Hindi, Punjabi, Korean, etc.).

Genre Categories – Organizes songs into genres such as Pop, Rock, Hip-Hop, Classical, Jazz, Indie, and EDM.

Artist Categories – Shows songs by different singers and bands.

Trending Songs – Displays the most popular and currently trending songs.

Search – Enables users to search songs using song title, artist, album, or keywords.

Login & Registration – Allows users to create an account and securely log in.

User Dashboard – Displays user profile information, favourite songs, and playlists (future implementation).

Contact Us – Provides a form through which users can submit feedback or queries.

2.3 Sub-Modules Description

Login/Signup: Allows users to create an account and securely log in to access personalized features.

Login/Signup: Enables users to create an account and securely log in. Registered users can access personalized features such as favourites and playlists.

Search Module: Allows users to quickly find songs using the song title, artist name, or keywords.

Music Categories: Helps users discover music through different categories such as Mood, Genre, and Language.

User Dashboard: Displays user profile details and favourite songs. Playlist management will be added in future versions.

Contact Us: Allows users to send feedback, suggestions, or report issues through an online contact form.

Functional Requirements

3.1. Software Requirements

HarmonyHub requires a web server capable of hosting a dynamic web application. The frontend will be developed using **HTML, CSS, and JavaScript** to provide an attractive and responsive user interface. **Node.js and Express.js** will be used for backend development, while **MySQL** will be used to manage user accounts, song information, favourites, and playlists. The website should be compatible with modern web browsers such as **Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari**.

3.2. Hardware Requirements

For the development of HarmonyHub, a computer with at least an **Intel Core i3 processor (or equivalent), 4 GB RAM (8 GB recommended), and 500 MB of available storage** is required. A stable internet connection is necessary for testing and deployment. Users can access the website using a desktop, laptop, tablet, or smartphone with any modern web browser and an internet connection.

3.3. User Requirements

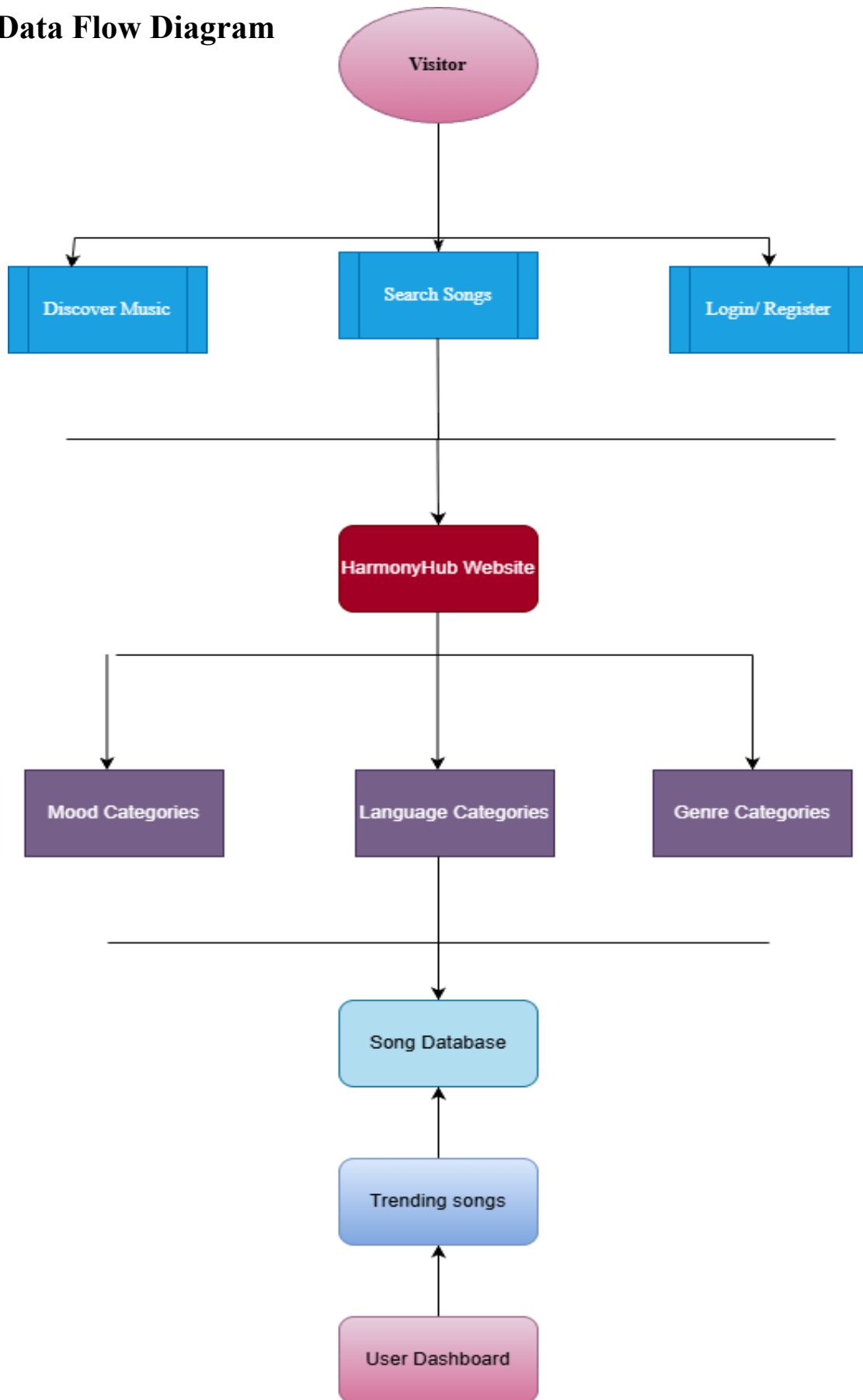
Users should be able to easily navigate the website and discover songs based on **mood, genre, language, artist, and trending categories**. They should also be able to search for songs, create an account, and log in to access personalized features. Registered users can manage their profile and favourite songs.

Table Description (Database Schema)

Module	Sub-Module	Description	User Type
Home	-----	Displays featured songs, trending tracks, and quick navigation.	Visitor, Registered User
Discover Music	-----	Allows users to explore the complete music collection.	Visitor, Registered User
Mood Categories	-----	Groups songs by mood such as Happy, Sad, Romantic, Party, and Relaxing.	Visitor, Registered User
Language Categories	-----	Organizes songs by language such as English, Hindi, Punjabi, and regional Indian languages.	Visitor, Registered User
Genre Categories	-----	Displays songs under genres like Pop, Rock, Hip-Hop, Indie, Classical, and EDM.	Visitor, Registered User
Artist Categories	-----	Shows songs by different singers and bands.	Visitor, Registered User
Trending songs	-----	Displays the most popular and currently trending songs.	Visitor, Registered User
Search	-----	Enables users to search songs using title, artist, or keywords.	Visitor, Registered User
Login and Registration	User Authentication	Allows users to create accounts and securely log in.	Visitor

User Dashboard	Favourite songs	Displays the user's saved favourite songs.	Registered User
Contact Us	Feedback Form	Allows users to submit feedback or enquiries.	Visitor, Registered User

Data Flow Diagram



References

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<https://developer.mozilla.org/en-US/docs/Web/HTML>
- MDN Web Docs – CSS
<https://developer.mozilla.org/en-US/docs/Web/CSS>
- MDN Web Docs – JavaScript
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